

CS 241 Data Organization using C

Linked Lists and Binary Trees

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This assignment is designed to let you work with linked lists and binary trees.

Problem specification

I have provided header files defining some functions related to linked lists and binary trees. I have also provided you with files to test these functions and a makefile to build it all.

Linked Lists

In a file named `linkedlist.c`, implement all the functions declared in `linkedlist.h` and make sure they work with `listtest.c`.

Binary Trees

In a file named `binarytree.c`, implement all the functions declared in `binarytree.h` and make sure they work with `treetest.c`.

1 Grading Rubric (total of 25 points + 3 bonus)

- 2 point** : The programs do not start with a comment stating the students first and last name and/or the source files are not named correctly.
- 5 points** : Programs compile with warnings on moons.cs.unm.edu using `/usr/bin/gcc` with the `-Wall -ansi -pedantic` options (using the makefile)
- 5 points** : Programs leak memory when tested with valgrind.
- 5 points** : All code follows the CS-241 standards (including comments).
- 10 points** : Output of running `listtest` matches `listtestout.txt`.

+1 bonus point : Don't create or destroy any nodes in the `reverseList` function. (This would also include indirectly creating/destroying nodes via the stack manipulation functions.) In other words, reverse the list in place by changing the pointers.

10 points : Output of running `treetest` matches `treetestout.txt`.

+1 bonus point : Only traverse the tree once in the `isBalanced` function. (The naive implementation using `maxDepth` does not meet this requirement.)

+1 bonus point : Only traverse the tree once in the `isBST` function.